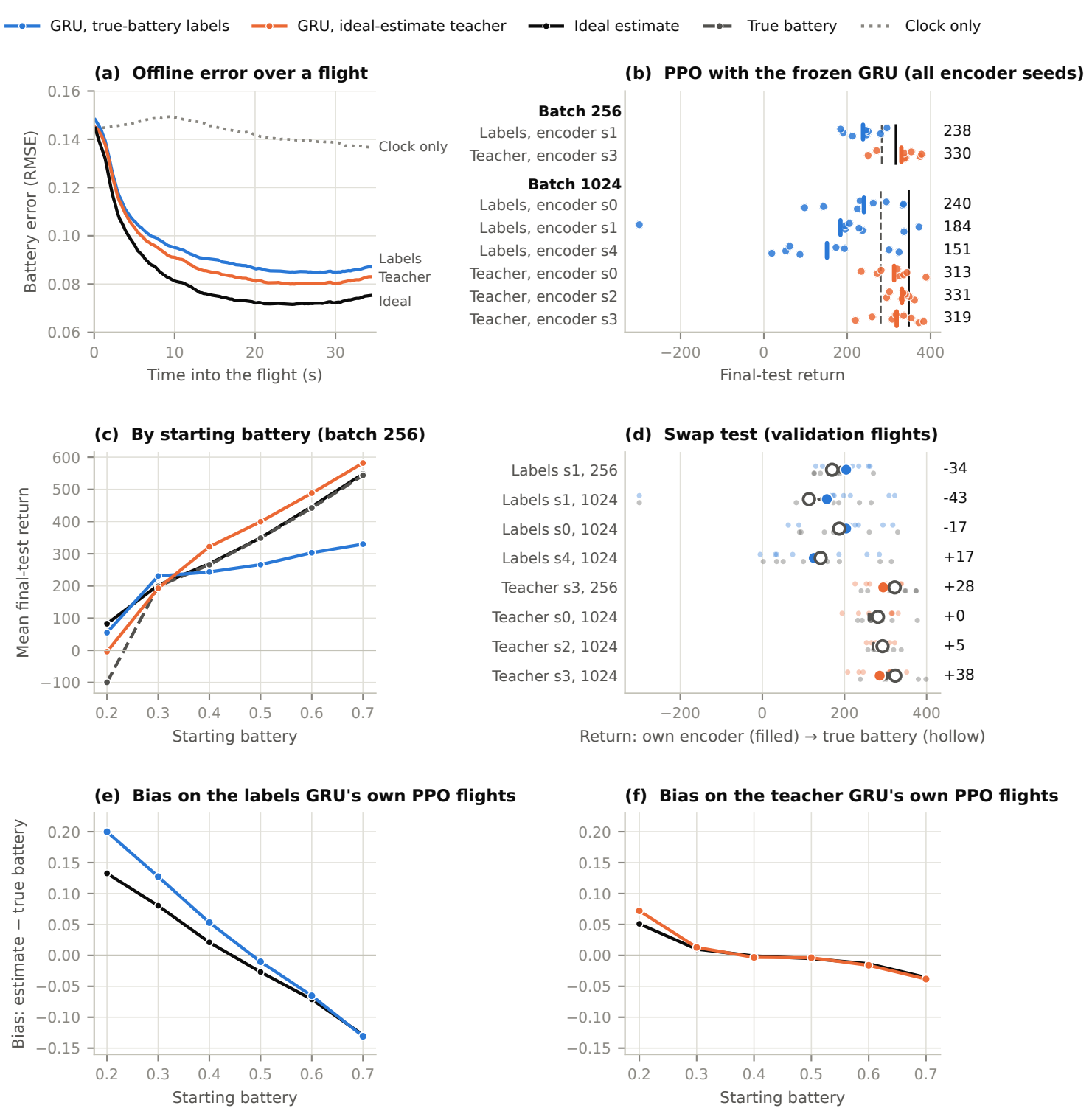

GRU, true-battery labels
GRU, ideal-estimate teacher
Ideal estimate
True battery
Clock only
(a) Offline error over a flight
Battery error (RMSE)
Time into the flight (s)
Clock only
Labels
Teacher
Ideal
(b) PPO with the frozen GRU (all encoder seeds)
Batch 256
Labels, encoder s1
Teacher, encoder s3
Batch 1024
Labels, encoder s0
Labels, encoder s1
Labels, encoder s4
Teacher, encoder s0
Teacher, encoder s2
Teacher, encoder s3
238
330
240
184
151
313
331
319
Final-test return
(c) By starting battery (batch 256)
Mean final-test return
Starting battery
(d) Swap test (validation flights)
Labels s1, 256
Labels s1, 1024
Labels s0, 1024
Labels s4, 1024
Teacher s3, 256
Teacher s0, 1024
Teacher s2, 1024
Teacher s3, 1024
-34
-43
-17
+17
+28
+0
+5
+38
Return: own encoder (filled) → true battery (hollow)
(e) Bias on the labels GRU's own PPO flights
Bias: estimate − true battery
Starting battery
(f) Bias on the teacher GRU's own PPO flights
Starting battery